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IC Basics and MOSFET

GATE ECE Electronic Devices and VLSI · Previous Year Questions with Answer Key

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Marks 2	40 questions
Answer key	at the end

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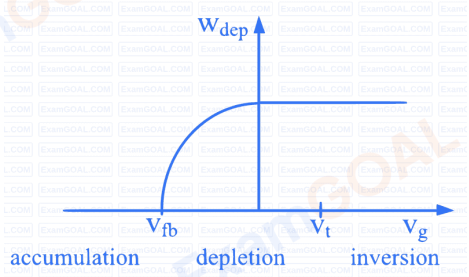
Marks 1 24 questions

Q1. GATE ECE 2023

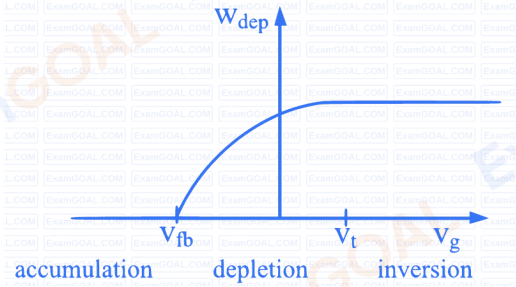
Open solution

For a MOS capacitor, V_{fb} and V_t are the flat-band voltage and the threshold voltage, respectively. The variation of the depletion width (W_{dep}) for varying gate voltage (V_g) is best represented by

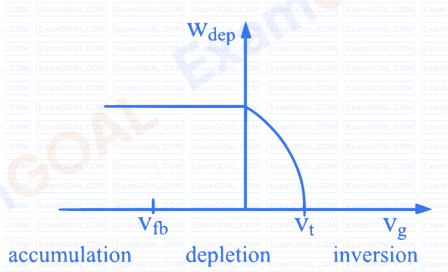
(A)



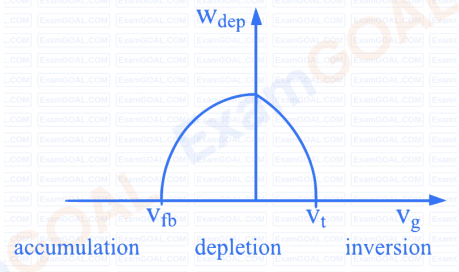
(B)



(C)



(D)



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Q2. GATE ECE 2017 Set 2

An n-channel enhancement mode MOSFET is biased at $V_{GS} > V_{TH}$ and $V_{DS} > (V_{GS} - V_{TH})$, where V_{GS} is the gate-to-source voltage, V_{DS} is the drain-to-source voltage and V_{TH} is the threshold voltage. Considering channel length modulation effect to be significant, the MOSFET behaves as a

Open solution

(A) voltage source with zero output impedance

(B) voltage source with non-zero output impedance

(C) current source with finite output impedance

(D) current source with infinite output impedance

Q3. GATE ECE 2017 Set 2

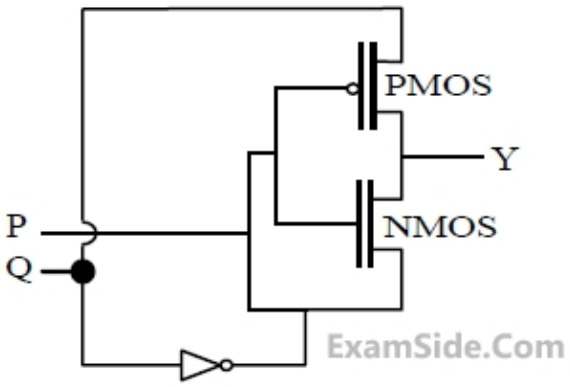
Consider an n-channel MOSFET having width W, length L, electron mobility in the channel μ_n and oxide capacitance per unit area C_{ox}. If gate-to-source voltage V_{GS}=0.7 V, drain-to source voltage V_{DS}=0.1V, $(\mu_n C_{ox}) = 100\ \mu A/V^2$, threshold voltage V_{TH}=0.3 V and (W/L) = 50, then the transconductance g_m (in mA/V) is _____.

Open solution

Q4. GATE ECE 2017 Set 2

For the circuit shown in the figure, P and Q are the inputs and Y is the output.

Open solution



The logic implemented by the circuit is

(A) XNOR (B) XOR (C) NOR (D) OR

Q5. GATE ECE 2016 Set 2

A long-channel NMOS transistor is biased in the linear region with V_{DS} = 50 mV and is used as a resistance. Which one of the following statements is NOT correct?

Open solution

(A) If the device width W is increased, the resistance decreases.

(B) If the threshold voltage is reduced, the resistance decreases.

(C) If the device length L is increased, the resistance increases.

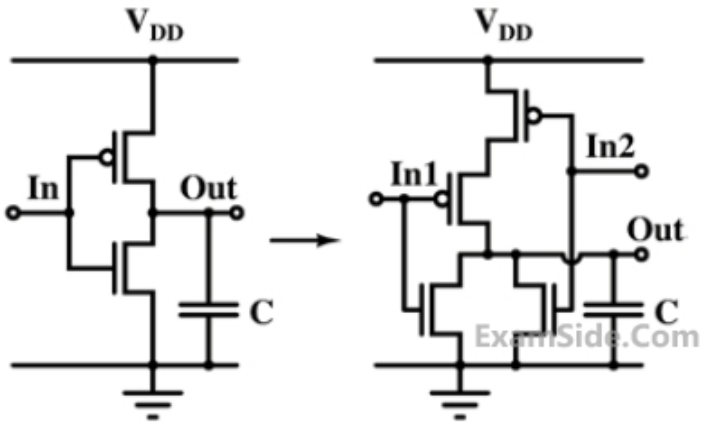
(D) If V_{GS} is increased, the resistance increases.

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Q6. GATE ECE 2016 Set 2

Open solution

Transistor geometries in a CMOS inverter have been adjusted to meet the requirement for worst case charge and discharge times for driving a load capacitor C. This design is to be converted to that of a NOR circuit in the same technology, so that its worst case charge and discharge times while driving the same capacitor are similar. The channel lengths of all transistors are to be kept unchanged. Which one of the following statements is correct?



(A) Widths of PMOS transistors should be doubled, while widths of NMOS transistors should be halved.

(B) Widths of PMOS transistors should be doubled, while widths of NMOS transistors should not be changed.

(C) Widths of PMOS transistors should be halved, while widths of NMOS transistors should not be changed.

(D) Widths of PMOS transistors should be unchanged, while widths of NMOS transistors should be halved.

Q7. GATE ECE 2016 Set 1

Open solution

Consider the following statements for a metal oxide semiconductor field effect transistor (MOSFET):

P: As channel length reduces, OFF-state current increases.
Q:As channel length reduces, output resistance increases.
R: As channel length reduces, threshold voltage remains constant.
S: As channel length reduces, ON current increases.
Which of the above statements are INCORRECT?

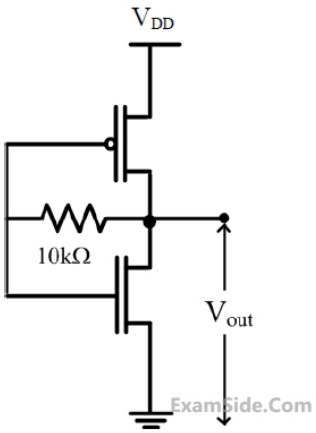
(A) P and Q (B) P and S (C) Q and R (D) R and S

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Q8. GATE ECE 2016 Set 1

Open solution

What is the voltage V_{out} in the following circuit?



(A) 0 V

(B) $(|V_T\ of\ PMOS|\ +\ V_T\ of\ NMOS)/2$

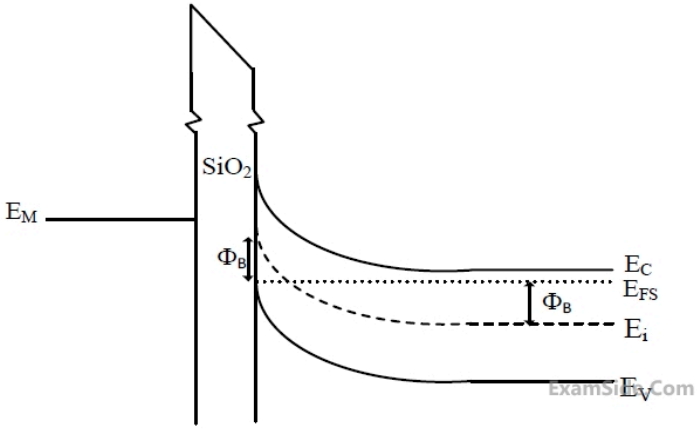
(C) Switching threshold of inverter

(D) V_{DD}

Q9. GATE ECE 2016 Set 3

Open solution

The figure shows the band diagram of a Metal Oxide Semiconductor (MOS). The surface region of this MOS is in



(A) inversion (B) accumulation (C) depletion (D) flat band

Q10. GATE ECE 2015 Set 3

Open solution

Which one of the following processes is preferred to from the gate dielectric (SiO_2) of MOSFETs?

(A) Sputtering

(B) Molecular beam epitaxy

(C) Wet oxidation

(D) Dry oxidation

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Q11. GATE ECE 2014 Set 1

If fixed positive charges are present in the gate oxide of an n-channel enhancement type MOSFET, it will lead to

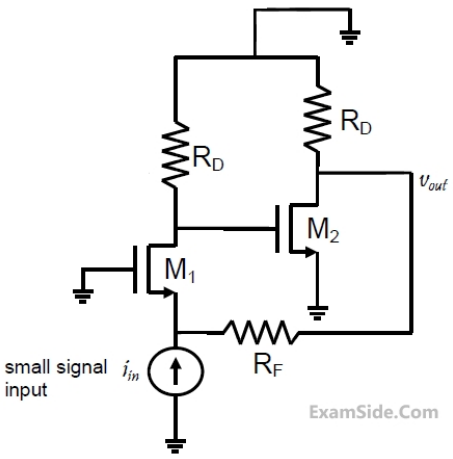
Open solution

(A) a decrease in the threshold voltage
(B) channel length modulation
(C) an increase in substrate leakage current
(D) an increase in accumulation capacitance

Q12. GATE ECE 2014 Set 1

In the ac equivalent circuit shown in the figure, if i_n is the input current and R_F is very large, the type of feedback

Open solution

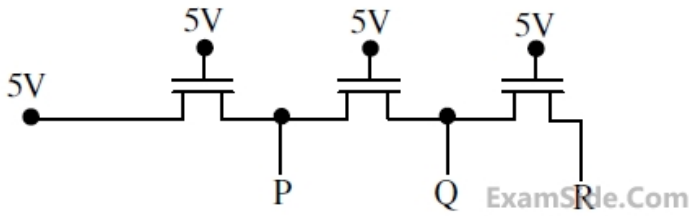


(A) voltage-voltage feedback
(B) voltage-current feedback
(C) current-voltage feedback
(D) current-current feedback

Q13. GATE ECE 2014 Set 1

In the following circuit employing pass transistor logic, all NMOS transistors are identical with a threshold voltage of 1 V. Ignoring the body-effect, the output voltages at P, Q and R are,

Open solution



(A) 4 V, 3 V, 2 V
(B) 5 V, 5 V, 5 V
(C) 4 V, 4 V, 4 V
(D) 5 V, 4 V, 3 V

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Q14. GATE ECE 2014 Set 2

In CMOS technology, shallow P-well or N-well regions can be formed using

(A) low pressure chemical vapour deposition
(B) low energy sputtering
(C) low temperature dry oxidation
(D) low energy ion-implantation

Open solution

Q15. GATE ECE 2014 Set 3

In MOSFET fabrication, the channel length is defined during the process of

(A) Isolation oxide growth
(B) Channel stop implantation
(C) Poly-silicon gate patterning
(D) Lithography step leading to the contact pads

Open solution

Q16. GATE ECE 2013

In IC technology, dry oxidation (using dry oxygen) as compared to wet oxidation (using steam or water vapor) produces

(A) superior quality oxide with a higher growth rate
(B) inferior quality oxide with a higher growth rate
(C) inferior quality oxide with a lower growth rate
(D) superior quality oxide with a lower growth rate

Open solution

Q17. GATE ECE 2013

In a MOSFET operating in the saturation region, the channel length modulation effect causes

(A) an increase in the gate-source capacitance
(B) a decrease in the Transconductance
(C) a decrease in the unity-gain cutoff frequency
(D) a decrease in the output resistance

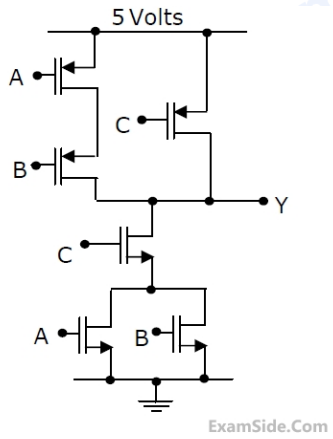
Open solution

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Q18. GATE ECE 2012

Open solution

In the circuit shown



(A) $Y = \overline{A}\,\overline{B} + \overline{C}$

(B) $Y = (A + B)C$

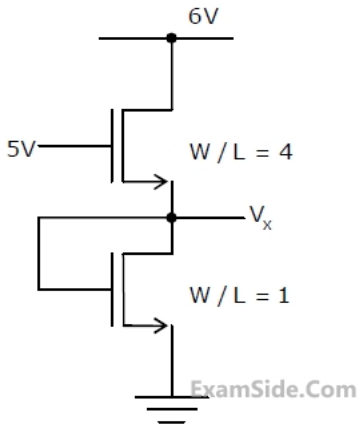
(C) $Y = \left(\overline{A} + \overline{B}\right)\overline{C}$

(D) $Y = AB + C$

Q19. GATE ECE 2011

Open solution

In the circuit shown below, for the MOS transistors, $\mu_n C_{ox} = 100\ \mu A/V^2$ and the threshold voltage V_T = 1 V. The voltage V_x at the source of the upper transistor is



(A) 1 V

(B) 2 V

(C) 3 V

(D) 3.67 V

Q20. GATE ECE 2008

Open solution

A silicon wafer has 100 mm of oxide on it and is inserted in a furnace at a temperature above 1000°C for further oxidation in dry oxygen. The oxidation rate

(A) is independent of current oxide thickness and temperature

(B) is independent of current oxide thickness but depends on temperature

(C) slows down as the oxide grows

(D) is zero as the existing oxide prevents further oxidation

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Q21. GATE ECE 2008

The drain current of a MOSFET in saturation is given by $I_D = K(V_{GS} - V_T)^2$ where 'K' is a constant. The magnitude of the transconductance g_m is

Open solution

(A) $\frac{K(V_{GS} - V_T)^2}{V_{DS}}$

(B) $2K(V_{GS} - V_T)$

(C) $\frac{I_d}{V_{GS} - V_{DS}}$

(D) $\frac{K(V_{GS} - V_T)^2}{V_{GS}}$

Q22. GATE ECE 2005

A MOS capacitor made using p-type substrate is in the accumulation mode. The dominant charge in the channel is due to the presence of

Open solution

(A) holes

(B) electrons

(C) positively charged ions

(D) negatively charged ions

Q23. GATE ECE 2004

Consider the following statements S1 and S2.

S1: The threshold voltage (V_T) of a MOS capacitor decreases with increase in gate oxide thickness

Open solution

S2: The threshold voltage (V_T) of a MOS capacitor decreases with increase in substrate doping concentration.

Which one of the following is correct?

(A) S1 is FALSE and S2 is TRUE

(B) both S1 and S2 are TRUE

(C) S1 is TRUE and S2 is FALSE

(D) both S1 and S2 are FALSE

Q24. GATE ECE 1994

The threshold voltage of an n-channel MOSFET can be increased by

Open solution

(A) increasing the channel dopant concentration

(B) reducing the channel dopant concentration

(C) reducing the gate-oxide thickness

(D) reducing the channel length

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Marks 2 40 questions

Q25. GATE ECE 2024

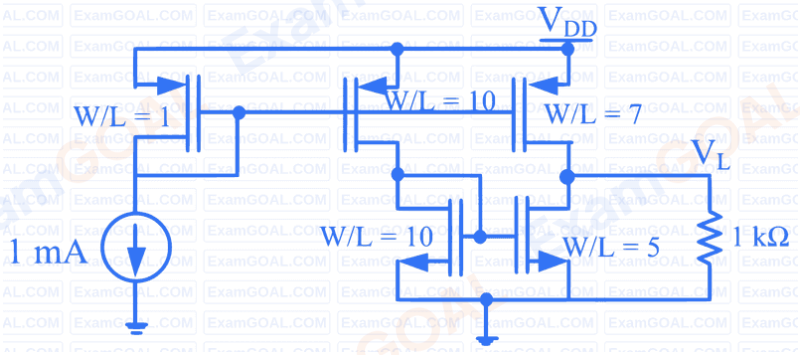
Open solution

Consider a MOS capacitor made with p-type silicon. It has an oxide thickness of 100 nm, a fixed positive oxide charge of 10^{-8} C/cm^2 at the oxide-silicon interface, and a metal work function of 4.6 eV. Assume that the relative permittivity of the oxide is 4 and the absolute permittivity of free space is 8.85×10^{-14} F/cm. If the flatband voltage is 0 V, the work function of the p-type silicon (in eV, rounded off to two decimal places) is _____.

Q26. GATE ECE 2023

Open solution

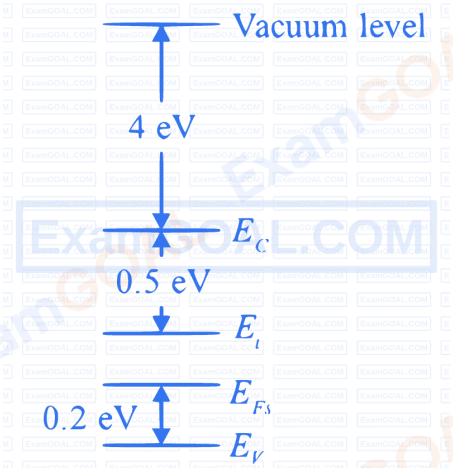
In the circuit below, the voltage V_L is _____ V (rounded off to two decimal places).



Q27. GATE ECE 2020

Open solution

The band diagram of p-type semiconductor with a bandgap of the 1 eV is shown. Using this semiconductor, a MOS capacitor having V_{TH} of -0.16 V, C'_{ox} of 100nF/cm^2 and metal work function of 3.87 eV is fabricated. There is no charge within the oxide. If the voltage across the capacitor is V_{TH}, the magnitude of depletion charge per unit area (in C/cm^2) is



(A) 1.41×10^{-8} (B) 0.52×10^{-8} (C) 0.93×10^{-8} (D) 1.70×10^{-8}

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Q28. GATE ECE 2017 Set 2

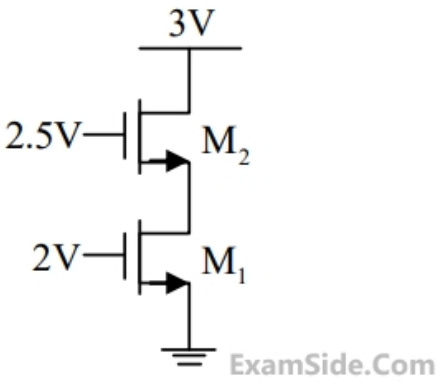
Open solution

A MOS capacitor is fabricated on p-type Si (silicon) where the metal work function is 4.1 eV and electron affinity of Si is 4.0 eV, E_c - E_F = 0.9 eV; where E_c and E_F are conduction band minimum and the Fermi energy levels of Si, respectively. Oxide$\varepsilon_r = 3.9,\ \varepsilon_0 = 8.85 \times 10^{-14}$ F/cm, oxide thickness $t_{ox} = 0.1\ \mu m$ and electronic charge q = 1.6×10^{-19} C. If the measured flat band voltage of the capacitor is −1V, then the magnitude of the fixed charge at the oxide semiconductor interface, in nC/cm^2, is _____.

Q29. GATE ECE 2017 Set 2

Open solution

Assuming that transistor M_1 and M_2 are identical and have a threshold voltage of 1V, the state of transistors M_1 and M_2 are respectively



(A) Saturation, Saturation
(B) Linear, Linear
(C) Linear, Saturation
(D) Saturation, Linear

Q30. GATE ECE 2017 Set 2

Open solution

Two n-channel MOSFETs, T1 and T2, are identical in all respects except that the width of T2 is double of T1. Both the transistor are biased in the saturation region of operation, but the gate overdrive voltage (V_{GS} - V_{TH}) of T2 is double that of T1, where V_{GS} and V_{TH} are the gate-to-source voltage and threshold voltage of the transistors, respectively. If the drain current and transconductance of T1 are I_{D1} and g_{m1} respectively ; the corresponding values of these two parameters for T2 are

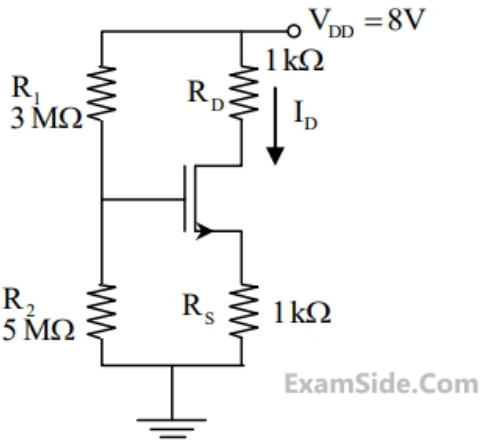
(A) $8I_{D1}$ and $2g_{m1}$
(B) $8I_{D1}$ and $4g_{m1}$
(C) $4I_{D1}$ and $4g_{m1}$
(D) $4I_{D1}$ and $2g_{m1}$

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Q31. GATE ECE 2017 Set 1

Open solution

For the circuit shown, assume that the NMOS transistor is in saturation. Its threshold voltage V_{tn} = 1 V and its trans-conductance parameter $\mu_n C_{ox}\left(\frac{W}{L}\right) = 1mA/V^2$. Neglect channel length modulation and body bias effects. Under these conditions the drain current I_D in mA is_____.



Q32. GATE ECE 2016 Set 2

Open solution

A voltage V_G is applied across a MOS capacitor with metal gate and p-type silicon substrate at T=300 K. The inversion carrier density (in number of carriers per unit area) for V_G = 0.8 V is $2 \times 10^{11}\ cm^{-2}$. For $V_G = 1.3\ V$, the inversion carrier density is $4 \times 10^{11}\ cm^{-2}$. What is the value of the inversion carrier density for V_G = 1.8 V?

(A) $4.5 \times 10^{11}\ cm^{-2}$
(B) $6.0 \times 10^{11}\ cm^{-2}$
(C) $7.2 \times 10^{11}\ cm^{-2}$
(D) $8.4 \times 10^{11}\ cm^{-2}$

Q33. GATE ECE 2016 Set 2

Open solution

Consider a long-channel NMOS transistor with source and body connected together. Assume that the electron mobility is independent of V_{GS} and V_{DS}. Given,

g_m = $0.5\mu A/V$ for V_{DS} = 50 m V and V_{GS} = 2V,

g_d = $8\mu A/V$ for V_{GS} = 2 V and V_{DS} = 0 V,

Where $g_m = \frac{\partial I_D}{\partial V_{GS}}$ and $g_d = \frac{\partial I_D}{\partial V_{DS}}$

The threshold voltage (in volts) of the transistor is

Q34. GATE ECE 2016 Set 1

Open solution

Consider an n-channel metal oxide semiconductor field effect transistor (MOSFET) with a gate-to-source voltage of 1.8 V. Assume that $\frac{W}{L} = 4, \mu_N C_{ox} = 70 \times 10^{-6} AV^{-2}$, the threshold voltage is 0.3V, and the channel length modulation parameter is 0.09 V^{-1}, In the saturation region, the drain conductance (in micro siemens) is_____.

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Q35. GATE ECE 2016 Set 3

Open solution

Figures I and II show two MOS capacitors of unit area. The capacitor in Figure I has insulator materials X (of thickness t_1 = 1 nm and dielectric constant ε_1 = 4) and Y (of thickness t_2 =3 nm and dielectric constant ε_2 = 200). The capacitor in Figure II has only insulator material X of thickness t_{eq}. If the capacitors are of equal capacitance, then the value of t_{eq} (in nm) is _____

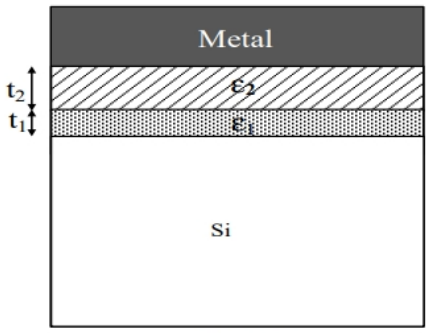


Figure I ExamSide.Com

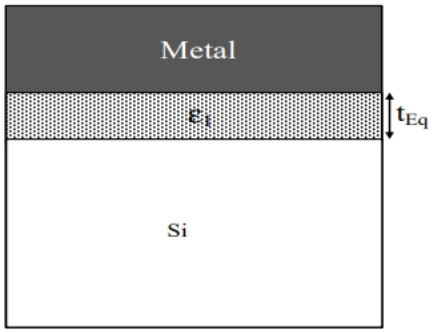


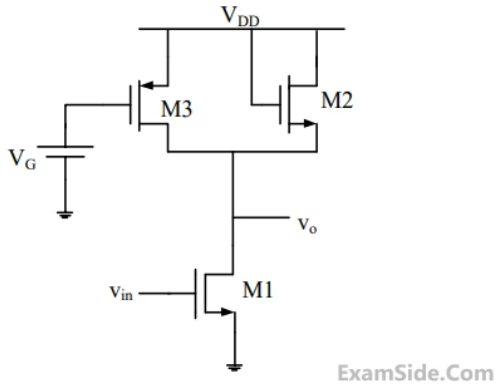
Figure II ExamSide.Com

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Q36. GATE ECE 2016 Set 3

Open solution

In the circuit shown in the figure, the channel length modulation of all transistors is non-zero $(\lambda \neq 0)$. Also all transistors operate in saturation and have negligible body effect. The ac small signal voltage gain (V_0/V_{in}) of the circuit is



(A) $-g_{m1}\left(r_{01}//r_{02}//r_{03}\right)$

(B) $-g_{m1}\left(r_{01}//\frac{1}{g_{m3}}//r_{03}\right)$

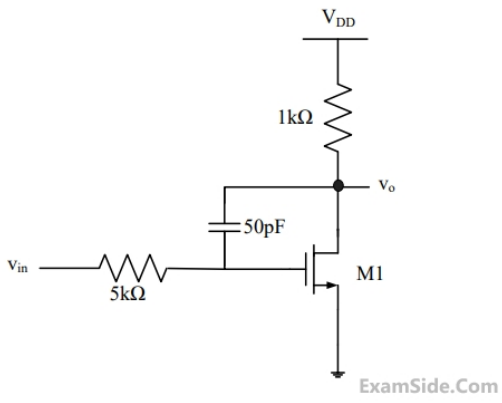
(C) $-g_{m1}\left(r_{01}//\left(\frac{1}{g_{m2}}//r_{02}\right)//r_{03}\right)$

(D) $-g_{m1}\left(r_{01}//\left(\frac{1}{g_{m3}}//r_{03}\right)//r_{02}\right)$

Q37. GATE ECE 2016 Set 3

Open solution

In the circuit shown in the figure, transistor M1 is in saturation and has transconductance g_m = 0.01 siemens. Ignoring internal parasitic capacitances and assuming the channel length modulation λ to be zero, the small signal input pole frequency (in kHz) is ____

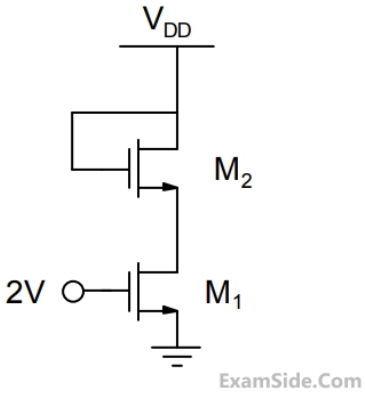


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Q38. GATE ECE 2015 Set 3

In the circuit shown, both the enhancement mode NMOS transistors have the following characteristics: $k_n = \mu_n C_{ox}(W/L) = 1mA/V^2$; V_{TN} = 1V. Asuume that the channel length modulation parameter λ is zero and body is shorted to source. The minimum supply voltage V_{DD} (in volts) needed to ensure that transistor M_1 operates in saturation mode of operation is _____

Open solution



Q39. GATE ECE 2015 Set 3

The current in an enhancement mode NMOS transistor biased in saturation mode was measured to be 1 mA at a drain-source voltage of 5 V. When the drain-source voltage was increased to 6 V while keeping gate-source voltage same, the drain current increased to 1.02 mA. Assume that drain to source saturation voltage is much smaller than the applied drain-source voltage. The channel length modulation p[arameter λ (in V^{-1}) is _____

Open solution

Q40. GATE ECE 2015 Set 2

In a MOS capacitor with an oxide layer thickness of 10 nm, the maximum depletion layer thickness is 100 nm. The permittivities of the semiconductor and the oxide layer are ε_s and ε_{os} respectively. Asuuming $\varepsilon_s/\varepsilon_{ox} = 3$, the ratio of the maximum capacitance to the minimum capacitance of this MOS capacitor is_____

Open solution

Q41. GATE ECE 2015 Set 1

A MOSFET in saturation has a drain current of 1 mA for V_{DS} =0.5V. If the channel length modulation coefficient is 0.05 V^{-1}, the output resistance (in kΩ) of the MOSFET is_____

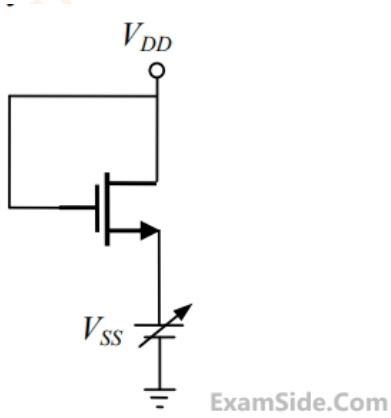
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Q42. GATE ECE 2015 Set 1

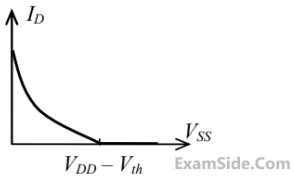
Open solution

For the NMOSFET in the circuit shown, in the threshold voltage is V_{th}, where $V_{th}>0$. The source voltage V_{ss} is varied from 0 to V_{DD}. Neglecting the channel length modulation, the drain current I_D as a function of V_{ss} is represented by



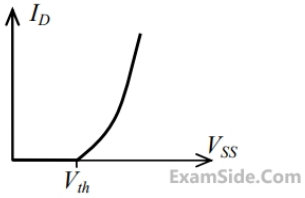
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(A)



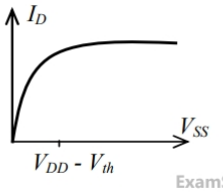
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(B)



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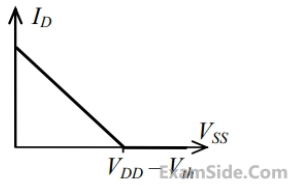
(C)



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(D)



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Q43. GATE ECE 2014 Set 1

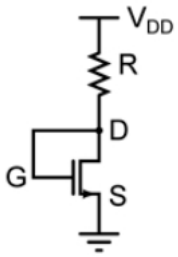
A depletion type N -channel MOSFET is biased in its linear region for use as a voltage controlled resistor. Assume threshold voltage V_{TH} = -0.5 V, V_{GS} = 2.0 V, V_{DS} = 5 V, W/L=100, $C_{OX}=10^{-8}$ F/cm^2 and μ_n = 800 cm^2/V-s. The value of the resistance of the voltage controlled resistor (in Ω) is _____.

Open solution

Q44. GATE ECE 2014 Set 2

For the n-channel MOS transistor shown in the figure, the threshold voltage V_{Th} is 0.8V. Neglect channel length modulation effects. When the drain voltage V_D = 1.6 V, the drain current I_D was found to be 0.5 mA.

Open solution



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If V_D is adjusted to be 2 V by changing the values of R and V_{DD}, the new value of I_D (in mA) is

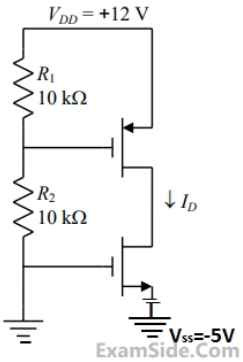
(A) 0.625 (B) 0.75 (C) 1.125 (D) 1.5

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Q45. GATE ECE 2014 Set 2

For the MOSFETs shown in the figure, the threshold voltage $|V_t|$ = 2V and K=$\frac{1}{2}\mu C_{OX}\left(\frac{W}{L}\right) = 0.1mA/V^2$. The value of I_D (in mA) is _____

Open solution



Q46. GATE ECE 2014 Set 3

The slope of the I_D vs. V_{GS} curve of an n-channel MOSFET in linear region is $10^{-3}\Omega^{-1}$ at V_{DS} = 0.1V. For the same device, neglecting channel length modulation, the slope of the $\sqrt{I_D}$ vs. V_{GS} curve (in $\sqrt{A}/V$) under saturation region is approximately _____.

Open solution

Q47. GATE ECE 2014 Set 3

An ideal MOS capacitor has boron doping concentration of 10^{15} cm^{-3} in the substrate. When a gate voltage is applied, a depletion region of width 0.5 μm is formed with a surface (channel) potential of 0.2V. Given that $\varepsilon_0 = 80854 \times 10^{-14} F/cm$ and the relative permittivities of silicon and silicon dioxide are 12 and 4, respectively, the peak electric field (in V/ μm) in the oxide region is _____

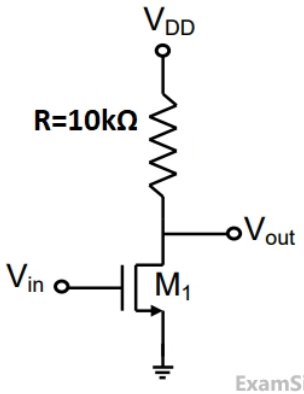
Open solution

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Q48. GATE ECE 2014 Set 3

For the MOSFET M_1 shown in the figure, assume W/L =2, V_{DD} = 2.0 V, $\mu n\ C_{OX}$ = 100$\mu A/V^2$ and V_{TH} =0.5 V. The transistor M_1 switches from saturation region to linear regionm when V_{in}(in Volts) is _____

Open solution

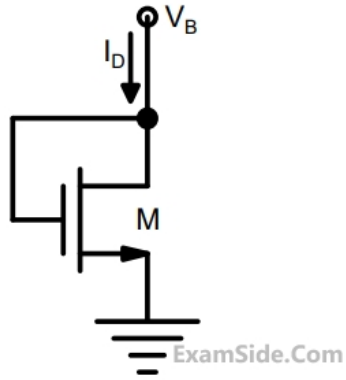


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Q49. GATE ECE 2013

The small-signal resistance (i.e., $\frac{dV_B}{dI_D}$) in $k\Omega$ offered by the n-channel MOSFET M shown in the figure below, at bias point of V_B = 2V is (device data for M: device transconductance parameter $k_N = \mu_n C'_{ox}$ (W/L)= 40$\mu A/V^2$, threshold voltage V_{TN}=1V, and neglect body effect and channel length modulation effects)

Open solution



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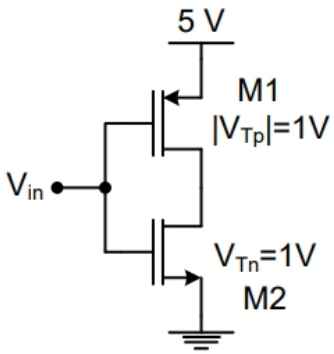
(A) 12.5 (B) 25 (C) 50 (D) 100

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Q50. GATE ECE 2012

Open solution

In the **CMOS** circuit shown, electron and hole mobilities are equal, and M_1 and M_2 are equally sized. The device M_1 is in the linear region if



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(A) $V_{in} < 1.875$ V

(B) $1.875 V < V_{in} < 3.125$ V

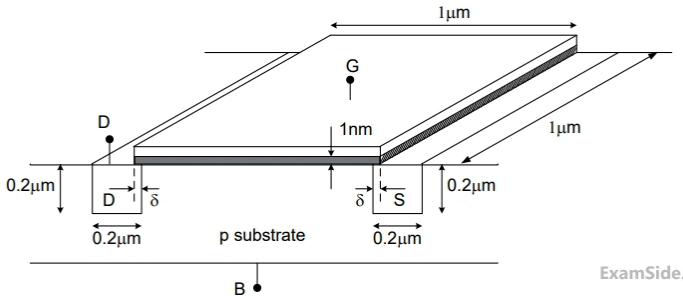
(C) $V_{in} > 3.125$ V

(D) $0 < V_{in} < 5$ V

Q51. GATE ECE 2012

Open solution

In the three dimensional view of a silicon n-channel **MOS** transistor shown below, $\delta = 20$ nm. The transistor is of width 1 μm. The depletion width formed at every p-n junction is 10 nm. The relative permittivities of Si and SiO_2, respectively, are 11.7 and 3.9, and ε_0 = 8.9 $\times 10^{-12}$ F/m.



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The source-body junction capacitance is approximately

(A) 2 fF

(B) 7 fF

(C) 2 pF

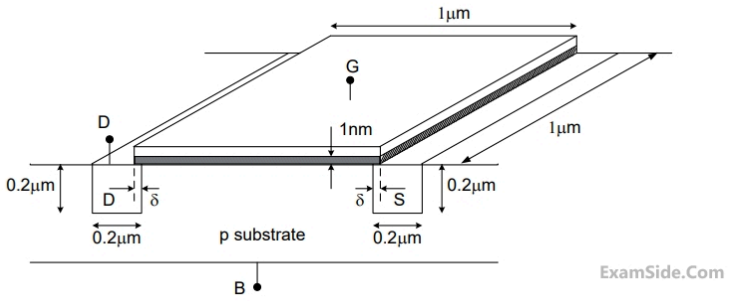
(D) 7 pF

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Q52. GATE ECE 2012

Open solution

In the three dimensional view of a silicon n-channel **MOS** transistor shown below, $\delta = 20$ nm. The transistor is of width 1 μm. The depletion width formed at every p-n junction is 10 nm. The relative permittivities of Si and SiO_2, respectively, are 11.7 and 3.9, and ε_0 = 8.9 $\times 10^{-12}$ F/m.



The gate-source overlap capacitance is approximately

(A) 0.7 fF (B) 0.7 pF (C) 0.35 fF (D) 0.24 pF

Q53. GATE ECE 2012

Open solution

The source of a silicon (n_i = 10^{10} per cm^3) n - channel **MOS** transistor has an area of 1 sq μm and a depth of 1 μm . If the dopant density in the source is $10^{19}/cm^3$, the number of holes in the source region with the above volume is approximately

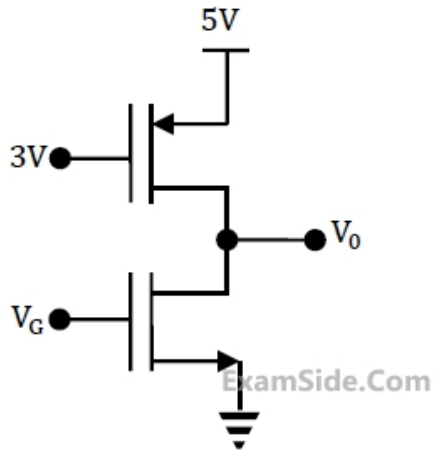
(A) 10^7 (B) 100 (C) 10 (D) 0

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Q54. GATE ECE 2009

Open solution

Consider the CMOS circuit shown, where the gate voltage of the n-MOSFET is increased from zero, while the gate voltage of the p-MOSFET is kept constant at 3 V. Assume that, for both transistors, the magnitude of the threshold voltage is 1 V and the product of the transconductance parameter and the $\left(\frac{W}{L}\right)$ ratio, i.e. the quantity $\mu C_{ox}\left(\frac{W}{L}\right)$, is 1 mAV^{-2}.



For small increase in V_G beyond 1 V, which of the following gives the correct description of the region of operation of each MOSFET?

(A) Both the MOSFETs are in saturation region

(B) Both the MOSFETs are in triode region

(C) n-MOSFET is triode and p-MOSFET is in saturation region

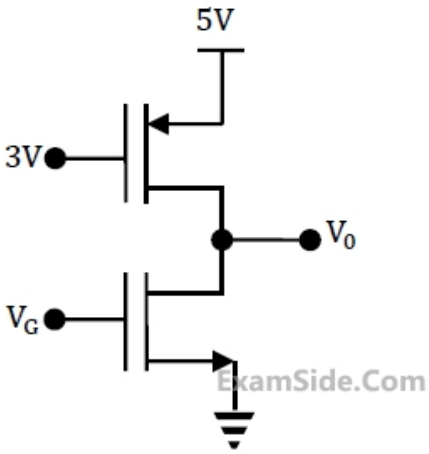
(D) n-MOSFET is in saturation and p-MOSFET is in triode region

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Q55. GATE ECE 2009

Open solution

Consider the CMOS circuit shown, where the gate voltage of the n-MOSFET is increased from zero, while the gate voltage of the p-MOSFET is kept constant at 3 V. Assume that, for both transistors, the magnitude of the threshold voltage is 1 V and the product of the transconductance parameter and the $\left(\frac{W}{L}\right)$ ratio, i.e. the quantity $\mu C_{ox}\left(\frac{W}{L}\right)$, is 1 mAV^{-2}.



Estimate the output voltage V_0 for V_G =1.5 V. [Hints: Use the appropriate current-voltage equation for each MOSFET, based on the answer]

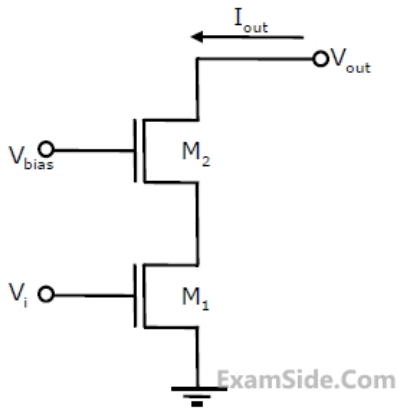
(A) $\left(4-\frac{1}{\sqrt{2}}\right)V$ (B) $\left(4+\frac{1}{\sqrt{2}}\right)V$ (C) $\left(4-\frac{\sqrt{3}}{2}\right)V$ (D) $\left(4+\frac{\sqrt{3}}{2}\right)V$

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Q56. GATE ECE 2008

Two identical NMOS transistors M_1 and M_2 are connected as shown below. V_{bias} is chosen so that both transistors are in saturation. The equivalent g_m of the pair is defined to be $\frac{\partial I_{out}}{\partial v_i}$ at constant V_{out}. The equivalent g_m of the pair is

Open solution



(A) The sum of individual g_m's of the transistors

(B) The product of individual g_m's of the transistors

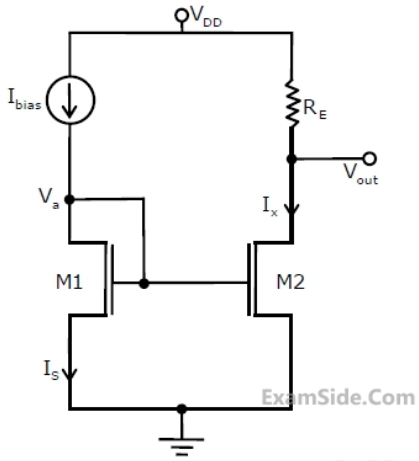
(C) Nearly equal to the g_m of M_1

(D) Nearly equal to g_m/g_0 of M_2

Q57. GATE ECE 2008

For the circuit shown in the following figure, transistors M1 and M2 are identical NMOS transistors. Assume that M2 is in saturation and the output is unloaded

Open solution



The current I_x is related to I_{bias} as

(A) $I_x = I_{bias} + I_s$

(B) $I_x = I_{bias}$

(C) $I_x = I_{bias} - I_s$

(D) $I_x = I_{bias}\left(V_{DD} - \frac{V_{out}}{R_E}\right)$

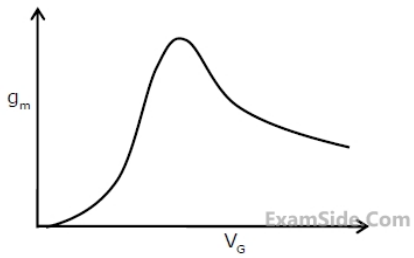
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Q58. GATE ECE 2008

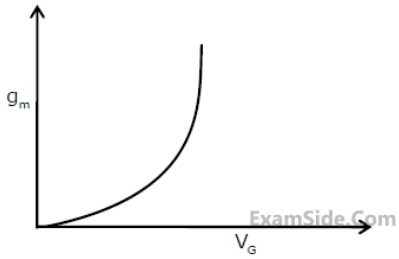
Open solution

The measured transconductance g_m of an NMOS transistor operating in the linear region is plotted against the gate voltage V_G at constant drain voltage V_D. Which of the following figures represents the expected dependence of g_m on V_G?

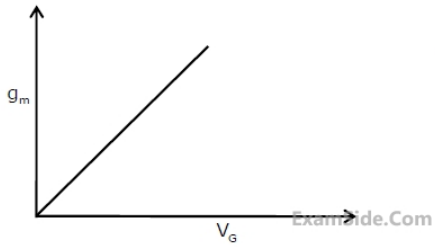
(A)



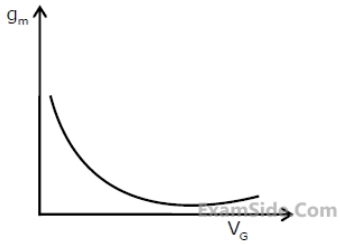
(B)



(C)



(D)



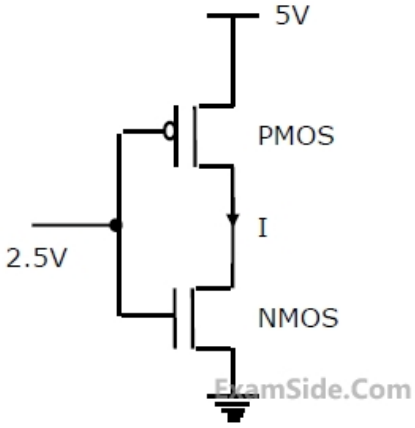
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Q59. GATE ECE 2007

Open solution

In the CMOS inverter circuit shown, if the transconductance parameters of the NMOS and PMOS transistors are

$K_n = K_p = \mu_n C_{OX} \frac{W_n}{L_n} = \mu_p C_{OX} \frac{W_P}{L_P} = 40\ \mu A/V^2$ and their threshold voltages are $V_T = 1$ V, the current I is:



(A) 0 A (B) 25 μA (C) 45 μA (D) 90 μA

Q60. GATE ECE 2006

Open solution

An n-channel depletion MOSFET has following two points on its $I_D - V_{GS}$ curve:

(i) $V_{GS} = 0$ at $I_D = 12$ mA and

(ii) $V_{GS} = -6$ Volts at $I_D = 0$

Which of the following Q-points will give the highest trans-conductance gain for small signals?

(A) $V_{GS} = -6$ Volts (B) $V_{GS} = -3$ Volts (C) $V_{GS} = 0$ Volts (D) $V_{GS} = 3$ Volts

Q61. GATE ECE 2004

Open solution

The drain of an n-channel MOSFET is shorted to the gate so that $V_{GS} = V_{DS}$. The threshold voltage (V_T) of MOSFET is 1 V. If the drain current (I_D) is 1 mA for $V_{GS} = 2$ V, then for $V_{GS} = 3$ V, I_D is

(A) 2 mA (B) 3 mA (C) 9 mA (D) 4 mA

Q62. GATE ECE 2003

Open solution

For an n-channel enhancement type MOSFET, if the source is connected at a higher potential than that of the bulk (i.e. $V_{SB} > 0$), the threshold voltage V_T of the MOSFET will

(A) remain unchanged (B) decrease

(C) change polarity (D) increase

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Q63. GATE ECE 2003

When the gate-to-source voltage (V_{GS}) of a MOSFET with threshold voltage of 400 mV, working in saturation is 900 mV, the drain current is observed to be 1 mA. Neglecting the channel width modulation effect and assuming that the MOSFET is operating at saturation, the drain current for an applied V_{GS} of 1400 mV is

Open solution

(A) 0.5 mA (B) 2.0 mA (C) 3.5 mA (D) 4.0 mA

Q64. GATE ECE 2003

If P is Passivation, Q is n-well implant, R is metallization and S is source/drain diffusion, then the order in which they are carried out in a standard n-well CMOS fabrication process, is

Open solution

(A) P-Q-R-S (B) Q-S-R-P (C) R-P-S-Q (D) S-R-Q-P

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Answer Key

Marks 1

1. **B**	2. **C**	3. **0.5**	4. **B**	5. **D**
6. **B**	7. **C**	8. **C**	9. **A**	10. **D**
11. **A**	12. **B**	13. **C**	14. **D**	15. **C**
16. **D**	17. **D**	18. **A**	19. **C**	20. **C**
21. **B**	22. **A**	23. **C**	24. **A**	

Marks 2

25. **4.10 to 4.50**	26. **2**	27. **D**	28. **6.9**	29. **C**
30. **B**	31. **2**	32. **B**	33. **1.2**	34. **28.35**
35. **1.6**	36. **C**	37. **57.87**	38. **3**	39. **0.022**
40. **4.33**	41. **20**	42. **A**	43. **500**	44. **C**
45. **0.9**	46. **0.0707**	47. **2.4**	48. **1.5**	49. **B**
50. **A**	51. **B**	52. **A**	53. **D**	54. **A**
55. **B**	56. **C**	57. **B**	58. **C**	59. **C**
60. **D**	61. **D**	62. **D**	63. **D**	64. **B**

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